Pyrite

△ **Pyrite crystals** in the form of pentagonal dodecahedrons, also called pyritohedrons

Known since antiquity, pyrite is better known by its informal title, “fool’s gold”. Its name is derived from the Greek word *pyr*, meaning “fire”, because pyrite emits sparks when struck by iron. Nodules of pyrite have been discovered in prehistoric burial mounds: the sun-like colour of pyrite probably assured its value. In later times, polished slices of its crystals were set edge to edge on wooden backing to make mirrors. Today, pyrite is polished as beads, and its bright crystals are themselves mounted as gemstones.

Specification

Chemical name Iron disulphide | **Formula** FeS_2
Colours Pale brass-yellow | **Structure** Cubic | **Hardness** 6–6.5
SG 5.0–5.2 | **RI** 1.81 | **Lustre** Metallic | **Streak** Greenish-black to brownish-black | **Locations** Spain, South America, USA, Japan, Italy, Norway, Greece, Slovakia

Spanish pyrite | **Rough** | The source of this specimen – Almira, Spain – is famous for its abundance of pyrite. These well-formed cubes are in a lime-rich mudstone matrix.

Pyrite crystal | **Rough** | This dazzling, neatly cuboid pyrite crystal offers a good demonstration of how the mineral can form in its natural state.

Modified crystals | **Rough** | The pyrite crystals in this excellent specimen have developed into the form of cubes modified by octahedrons.

Pyrite necklace | **Set** | The spherical beads of this necklace are made of highly polished pyrite, finely crafted even though pyrite is brittle and difficult to work.

Pyrite and quartz | **Rough** | In this classic pyrite specimen, prismatic quartz crystals are growing on octahedral crystals of pyrite. The two often grow together.

Marcasite

Pyrite in disguise

Marcasite is a mineral that, in all likelihood, has never been used as a gemstone. However, the name is widely used to refer to both it and pyrite. So-called marcasite jewellery, popular in Victorian times, has always been made mainly from pyrite, since some genuine marcasite is chemically unstable and rapidly deteriorates in air. Although marcasite is chemically identical to pyrite, it has a different crystal structure.

Cut “marcasites” This mass of rose-cut “marcasites” – actually pyrite – are ready for setting.